O-ETB Evidence Repo and the Evidence Manager

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Introduction

This is a working document last updated on the date shown in the page footer.

This document provided information related to the use of external evidence-generating tools/processes with the O-ETB assurance tool. It has been proposed to have a central repository for evidences. Previously referred to as “centralDB” or “Evidence Manager”, to be used in connection with the O-ETB Assurance Tool for the construction and review of assurance cases, on the MedSecurance Project. The Evidence Manager was proposed by Miroslaw “Mirek” Malinowski of University of Warwick as a lightweight database for storing all evidence generated by various MedSecurance project-developed or other third-party tools that supports assurance argumentation presented in assurance cases generated by the O-ETB Assurance Tool.

We seek to identify issues with the features and interactions of these components and to find solutions. This is necessary to determine the features needed of the Evidence Manager and changes necessary to current features of the O-ETB Assurance Tool.

We begin by reciting some facts about the O-ETB and its operation and then turn to its potential interaction with the Evidence Manager and open issues. Hopefully, this discussion will help us to better understand and find solutions to the issues.

O-ETB Repositories

The ETB maintains two repositories that it creates while generating an assurance case (AC):

- the EVIDENCE repository, and
- the CASES repository

both contained in the top-level directory MedSecurance/Assurance/REPOSITORY. The repositories are dynamically created, maintained and used by the O-ETB Assurance Tool directly and, in the case of the EVIDENCE repository, updated by “tool agents” that are invoked by the Assurance Tool.

EVIDENCE Repository

Each tool agent is associated with a particular evidence category, providing a wrapper for the invocation of “evidential tools” or “evidential processes” by O-ETB. These are automated tools or semi-automated processes that provide concrete evidence instances upon completion of the procedures that they embody. Tool agents are trusted by O-ETB to enforce the evidence management protocol of the O-ETB while marshaling the needed inputs for the corresponding tool, executing the tool, gathering and interpreting the tool execution results and appropriately updating the record corresponding to the particular instance in the EVIDENCE repository. After invocation the tool agent executes and updates the repository asynchronously. See also O-ETB User Manual, Section 3.8.4

An ac_evidence record in the EVIDENCE repository has the following elements:

- category : an atom belonging to an association between defined atoms and evidence categories
- claim : a text string that is the claim being supported by the evidence
- context : a list of text strings that characterize the context
- aargs : a list of items passed by the assurance case instantiation procedure that supply information needed by the corresponding tool agent, including arguments to the evidential tool
- xref : a positive integer that constitutes a unique identifier for the evidence record
- status : a symbol that is one of the set {pending, ongoing, valid, invalid} representing the state of the evidence record

CASES Repository

The CASES repository contains an internal form that represents the incremental construction of the assurance cases stored in the repository under a name provided by the user when the AC is first instantiated. Generally the user does not need to interact with this representation so we do not detail it further here.

O-ETB and collaborating tools

The O-ETB Core Engine, using information in its Knowledge Base and driven by user commands (either interactively or by command scripts), builds an assurance case in the CASES Repository with its supporting evidence contained in the EVIDENCE Repository. For each item of evidence required by the argumentation of the assurance case the Core Engine creates a placeholder record according to the evidence category and invokes the corresponding tool agent as defined in the Knowledge Base. The tool agent for an evidence category is invoked by the Core Engine according to an invocation protocol defined by O-ETB. The tool agent embodies the specific knowledge of how to invoke the corresponding tool and interpret its ouput. Figure 1 shows several organizational patterns of external tools in relation to their corresponding tool agent.

A special kind of agent (such as Agent D in the figure) interfaces the O-ETB to external assurance processes that partially or completely rely on human activity. The result of such processes are recorded as updates to corresponding evidence records in the EVIDENCE Repository in the same manner that Tool Agents make such updates to record the results of running external evidential tools.

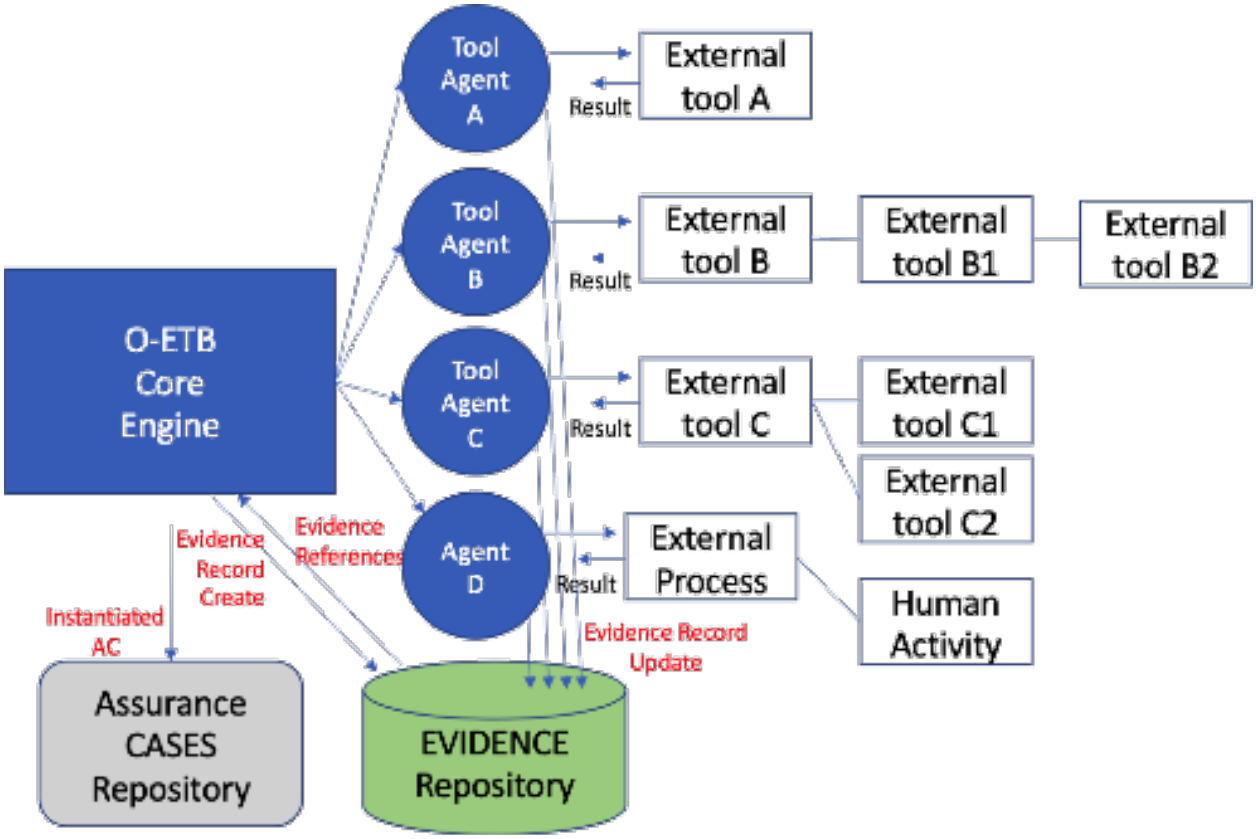

Figure 1 - O-ETB Core, Repositories and Tool Agents

Event sequence of O-ETB and REPOSITORY actions

The previous version (v1) of the O-ETB sequence diagram (Figure 2) showed only the entities involved in the O-ETB Assurance Case Generation process. It did not include the pre-generation assurance engineering activity or the post-generation activity of exporting and reviewing the exported assurance case.

A new version (v2) of the sequence diagram depicts all three activities. Because v1 was at the limit of what could be legibly included as a figure in this document, v2 is supplied in a separate draw.io source file and its corresponding PDF file. Please refer to the v2 (or subsequent latest) file for this discussion.

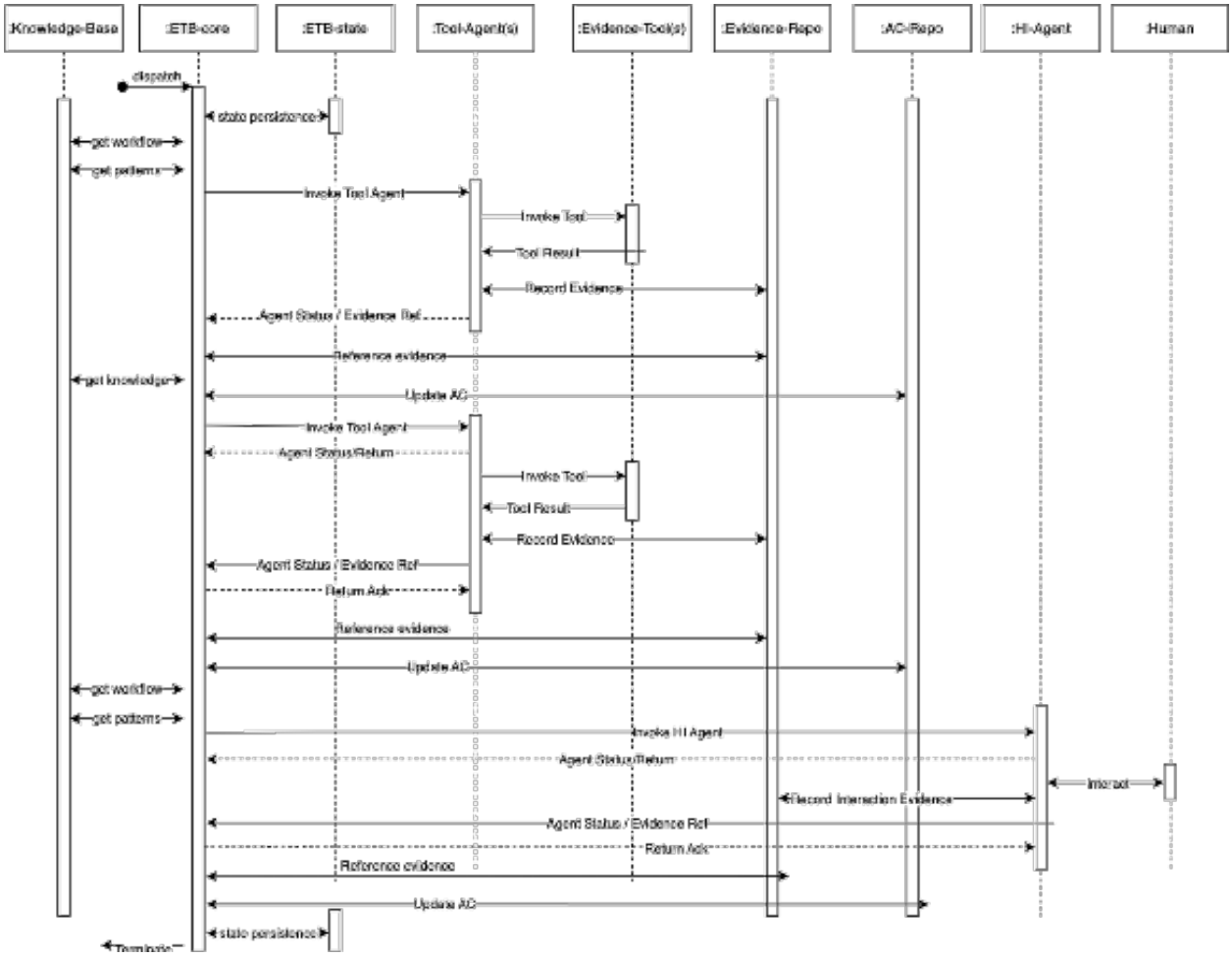

Figure 2 - O-ETB sequence chart v1

In v2 the sequence diagram is divided vertically into three phases: Pre-Generation Phase I, Generation Phase II, and Post-Generation Phase III. These phases comprise respectively the Assurance Engineering Activity, The Assurance Case Generation Process, and the Assurance Case Review Activity. Communications among entities in the diagram that are related to the Assurance Engineering Activity are shown in green and those of the Assurance Case Review Activity are in orange. Those that are part of the Assurance Case Generation Process are shown in black as in v1.

The two new Phases and their corresponding activities are depicted in a minimal way currently in v2. It is our intention that further details of the interactions, particularly with the proposed CentralDB by the Evidence Tools and the O-ETB will be elaborated with the collaboration with the other evidence tool development partners.

Evidence Manager concept

A central repository for evidence files implemented as a simple database mechanism providing for storing, retrieving, listing, and deleting evidence that The Evidence Manager is fronted by a simple API gateway. It would have built-in version control and the ability to attach comments.

Questions and issues related to the EVIDENCE Repo and the CentralDB

In order to harmonize the concepts of the O-ETB, in particular its EVIDENCE Repository, with the external evidence tools and the proposed CentralDB or Evidence Manager, it is essential that the role and structure O-ETB's EVIDENCE Repository and its relation to the operation of O-ETB is well understood, so that requirements on the CentralDB and O-ETB can be clearly expressed. We suggest that we characterize the usage and interaction scenarios and collaboratively flesh-out the v2 sequence diagram with respect to the interactions of the Evidence Manager with the O-ETB components. To that end, following are several facts and questions about the O-ETB and the CentralDB:

1. As noted previously, the EVIDENCE repo contains ac_evidence records comprising the following fields: category, claim, context, aargs, xref, and status.
2. The xref field is a semantically void identifier that is unique to the record within the EVIDENCE repo and serves as a compact reference that can be used to efficiently retrieve the other fields of the record.
3. The four initial fields of the record, k = (category, claim, context, aargs) represent a semantically rich signifier of a piece of evidence. That is the definitions in the KB of the evidence categories and implementation of the tool agents along with the semantics of the tools invoked by the tool agents together form a function evidence() which applied to k as in evidence(k), where k implicitly or explicitly provides the identity other artifacts concerning the target of assurance (TOA) that are needed to evaluate the evidence function, returns the facts comprising the evidence.
4. Note that the validity, value and sufficiency of the evidence item is strictly interpreted with respect to the context of its usage and the argument structure into which it is incorporated. Any nuance on the interpretation and applicability of the evidence item has been predetermined by the designers of the argument structure, or of the general argument patterns used construct the specific argument structure.
5. The assurance case structure and the appearance of evidence (solution) nodes therein determine that specific parameters (k) for a suitable evidence item.
6. The assurance case argument pattern library used is like the body of a program. The O-ETB Core Engine's instantiate function is the interpreter. The invocation of the instantiate function with an identified top-level pattern (entry point to the program) and the actual parameters supplied to satisfy the formal parameters of the specified pattern, initiates the assurance case generation process.
7. The execution of the AC generation generates in turn the specific subgoals of the assurance argument and their corresponding evidential demands. Changes to assurance strategy or tactics are implemented by changes to the pattern library or to the instantiation invocation parameters (which may include changes to specification of the

TOA). Such changes may result in novel specific evidence function invocations. It is not in general practical to pre-generate all the evidence items that may be demanded as the top-level assurance strategy and tactics are adapted.

8. With the possible exception of analyses performed by tools that may be applicable to many variations of assurance strategy and tactics, re-generation of an assurance case will create novel specific evidence item demands. This agrees with the observation that there is no need to re-run an assurance case generation unless something among the inputs, or the tools, or how other assurance activities are performed has been changed.
9. This leads to the questions: What evidence items are contained in the Evidence Manager? What determines what evidence items are contained there? How is this determination related to the evidence demands arising from assurance case generation? What metadata are associated with the evidence items? If there is an evidence manager outside of O-ETB how are its contents kept in sync with the O-ETB EVIDENCE Repository over multiple runs?

Usage roles for the EVIDENCE Repository and Evidence Manager

Several usage roles can be described to elaborate on the activities presented in the sequence diagram v2. The currently identified roles are:

- Assurance engineer role
- Assurance tool developer
- Evidential tool developer role
- Assurance user role
- Assurance reviewer role

These will be described in turn. Note that each of these may subsequently be refined into multiple roles. In the following “TOA” stands for Target of Assurance, the product, system, tool or method for which the assurance case is being developed.

Assurance engineer role

The assurance engineer role takes into account the strategy and tactics of assurance. The role identifies the scope of its current effort and develops and tests assurance methodologies on examples of the intended scope of application demonstrating its efficacy. The role includes the identification and adoption of assurance processes and tools to achieve the assurance aims within the domain. This includes choosing or recommending already existing tools and processes as well as specifying the requirements for new evidential or process tools to serve the assurance strategy. The outputs include guidance on the use of tools and processes as well as codifying assurance case argument patterns, evidence categories, tools and processes (that generate the identified evidence categories) within the Assurance Tool Knowledge Base for goals that are representative of the domain and the anticipated kinds of TOA.

The assurance engineer role should determine whether the assurance cases and other information exported by the Assurance Tool should be rendered in a particular, currently unsupported format, for consumption by the assurance reviewer role. The role should

communicate the requirements for such renderings and collaborate on their implementation by the Assurance Tool developer and the Evidential tool developer roles.

Assurance Tool developer role

The Assurance Tool developer (currently The Open Group) develops and tests the assurance tool which generates assurance cases for any TOA based on a Knowledge Base (KB). The KB includes knowledge provided by the other roles that is expressed domain-specific languages (DSLs) that are defined by the assurance tool. This may include both generic knowledge and knowledge specific to a particular application domain, specific to the intended certification regime, or even specific to the TOA. The assurance tool developer role includes making adaptations to the assurance tool to accommodate new concepts and data types, and to extend the DSLs to accommodate these novel or domain-specific entities. Finally, this role supports the development of new renderings of the assurance cases according to the requirements of specific certification regimes as recommended by the assurance engineer role.

Evidential tool developer role

The evidential tool developer develops new tools, or adapts existing tools, to support the assurance methodologies by generating instances of evidential artifacts that will be incorporated into assurance argumentation in support of assurance goals. The activity includes integrating the new or adapted tools into established automation structures so that its generated evidential artifacts can be incorporated in a uniform way. In the case of the Assurance Tool used on the MedSecurance Project this includes defining the evidence category generated by each tool, demonstrating how the evidence category is incorporated into assurance case patterns, and creating the wrapper for the tool, known as a “tool agent,” that enables the tool to be invoked by the Assurance Tool and the results of executing the tool to be incorporated into the operation of the Assurance Tool and its resulting assurance cases.

The evidential tool developer role should determine how the input artifacts and generated evidence artifacts are to be presented to representatives of the assurance reviewer role.

Assurance user role

The assurance user role employs the available assurance methodologies, assurance tool, argument patterns, knowledge base, assurance activities, evidential tools and associated tool agents to carry out an assurance strategy for a specific TOA. The assurance user role primarily uses existing methods and tools but may discover the need to customize or tailor existing items for use within a specific application case. Such customization may include modifications to items in the Assurance Tool Knowledge Base, which the assurance user is expected to do (potentially with guidance), or adaptations to the tools which should be pursued in collaboration with the Assurance and Evidential tool developers.

The assurance user role should determine that the outputs of the Assurance and Evidential tools conform to the requirements of the target certification regime and to the assurance

reviewer role. If there is some inadequacy for the assurance users' aims there should be collaboration with the tool developer roles.

Assurance reviewer role

The assurance reviewer role may be carried out within a project by a representative of a TOA originator or a representative of a certification body. The in-project review is a critical assessment taken from the viewpoint of the certifier to provide confidence that the body of assurance evidence presented is consistent with the expectations of the certifier to avoid exceptions being raised during the certifier's assurance review.

References